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Scenario - based

CO2 - AT2

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2. Point detection

I_b is calculated by Laplacian operator

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$R = \sum w_i z_i$$

$$\text{Mask} = \begin{bmatrix} -1 & -1 & -1 \\ -1 & 8 & -1 \\ -1 & -1 & -1 \end{bmatrix}$$

→ Horizontal line means

image:

$$\begin{bmatrix} 45 & 48 & 46 \\ 47 & 220 & 49 \\ 46 & 48 & 49 \end{bmatrix}$$

$$T = 500$$

Step 1: Multiply with mask and img value.

$$R = -45 - 48 - 46 - 47 - 49 - 46 - 48 - 47 + 1760 = 1760 - 376$$

$$R = 1384$$

threshold value = 500

There is white point in a image condition;

$$R > T$$

$$1384 > 500$$

→ We found one white dot in tile image. So, the tile is rejected. That is called point detection.

a) Point detection response gives in the "r" value is 1384. Since the threshold value is 500.

b) The "r" value is greater than 500 so, there is point detected.

Line detection:

$$R = W \times f(x, y)$$

i) Horizontal line $\Rightarrow$ mask =
$$\begin{bmatrix} -1 & -1 & -1 \\ 2 & 2 & 2 \\ -1 & -1 & -1 \end{bmatrix}$$

ii) Vertical line $\Rightarrow$ mask =
$$\begin{bmatrix} -1 & 2 & -1 \\ -1 & 2 & -1 \\ -1 & 2 & -1 \end{bmatrix}$$

iii) +45° Diagonal line $\Rightarrow$ mask =
$$\begin{bmatrix} 2 & -1 & -1 \\ -1 & 2 & -1 \\ -1 & -1 & 2 \end{bmatrix}$$

iv) negative diagonal line $\Rightarrow$ mask =
$$\begin{bmatrix} -1 & -1 & 2 \\ -1 & 2 & -1 \\ 2 & -1 & -1 \end{bmatrix}$$

To identify line, $R = \sum_{i=-1}^1 \sum_{j=-1}^1 W(i, j) f(x+i, y+j)$

2) Solution:

$$\text{image} = \begin{bmatrix} 20 & 20 & 20 \\ 100 & 100 & 100 \\ 20 & 20 & 20 \end{bmatrix}$$

$$\begin{aligned} \text{Horizontal line} &= -20 - 20 - 20 - 20 - 20 - 20 + 200 + 400 \\ &= 600 - 120 \\ &= 480 \end{aligned}$$

$$\text{Vertical line} = -20 + 40 - 20 + 100 + 200 - 100$$

$\frac{1}{2} \times 10 \times 10 = 50$

1. $\frac{1}{2} \times \frac{1}{2} = \frac{1}{4}$

(8, 30) $\theta = 1.1 = 1$
orientation is

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1. 1000 (4 mil)

Canny Edge detector:

edge edge detect

Agavea villosa (L.)

- 2 — and probably 10

ii) non-maximum

iv) Double thresholding

hysteresis

100 100 100

$$f(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{x^2 + y^2}{2\sigma^2}}$$

$$= \sqrt{G_x^2 + G_y^2}$$

strong edge pinch.

step: 4: Two thresholding values are used T_{high} & T_{low}
Pixel is classified as

- Strong edge
- Weak edge
- Non-edge

3. solution:

Horizontal Gradient (G_x) = 36

Vertical Gradient (G_y) = 48

Previous pixel = 42

center pixel = 60

Next pixel = 35

High threshold = 55

Low threshold = 30

a) To remove noise.

b)

$$G = \sqrt{G_x^2 + G_y^2}$$
$$= \sqrt{36^2 + 48^2}$$
$$= \sqrt{3600}$$

$$G = 60$$

c) No,

$42 < 55$: Weak edge

$60 > 55$: Strong edge

$35 < 55$: Weak edge

d)

Previous pixel - Weak edge

center pixel - Strong edge

Next pixel - Weak edge

e) yes.

f) Canny detection algorithm is very much powerful in edge detection.

Edge linking and boundary Problems:

Edge is a location, where img intensity changes significantly.

Types:

- i) Step edge
- ii) Ramp edge
- iii) Roof edge
- iv) Line edge

$$\Delta f(x, y) = \frac{\partial f}{\partial x} + \frac{\partial f}{\partial y}$$

$$|\Delta f| = \sqrt{G_x^2 + G_y^2}$$

Edge direction:

- Horizontal (0°)
- Vertical (90°)
- Diagonal (45°)
- Anti diagonal (135°)

$$\text{Gradient direction} = \theta = \tan^{-1} \left(\frac{G_y}{G_x} \right)$$

Steps:

1. select an edge pixel.
2. Examine neighbouring pixel.
3. compare gradient properties.
4. connect similar pixel.
5. Repeat until boundary is formed.

Hough transform:

It is a global feature extraction technique used to detect geometric shapes such as lines and circle from edge pixels.

Q4) A self driving car uses a camera using a lane to detect lane marking as a highway after apply the canny edge detection

$$P_1 = 10, \theta = 44^\circ$$

$$P_4 = 18, \theta = 92^\circ$$

$$P_2 = 11, \theta = 46^\circ$$

$$P_5 = 19, \theta = 91^\circ$$

$$P_3 = 12, \theta = 45^\circ$$

threshold value is less than 2. $G_x = 44$, $G_y = 91$

Sol:

$$G = \sqrt{44^2 + 91^2}$$

$$= 101.07$$

$$\approx 101$$

Step 1: Find distance b/w P_1 and P_2

$$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$$

$$= \sqrt{(11 - 10)^2 + (11 - 10)^2}$$

$$= \sqrt{2} = 1.414$$

Gradient direction, linking condition

$$|G_1 - G_2| < T_g \rightarrow \text{Gradient magnitude}$$

$$|\theta_1 - \theta_2| < T_\theta$$

$$|46 - 44| < 5$$

$$2 < 5$$

$\therefore$ Two edges are linked.

Blur P_2 and P_3

$$d = \sqrt{2} = 1.414$$

Linking condition:

$$GM: 1 < 2$$

$$GD: 1 < 5$$

$\therefore$ Linked.

Blur P_3 and P_4

$$d = \sqrt{32} = 4\sqrt{2}$$

$$= 8.485$$

Linking condition

$$GM: 6 \nless 5$$

$$GD: 47 \nless 2$$

$\therefore$ It is not linked.

Blur P_4 and P_5

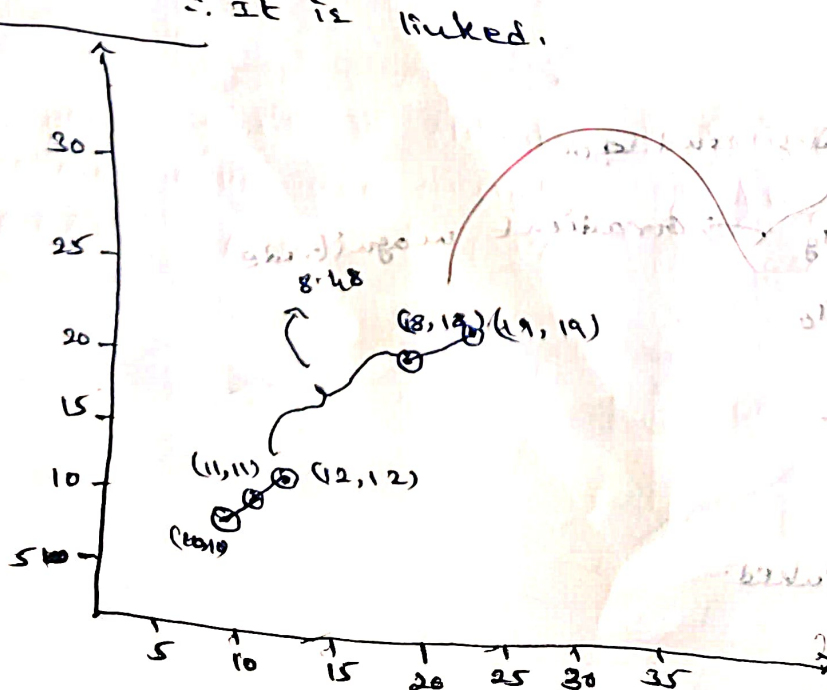
$$d = \sqrt{2} = 1.414$$

Linking condition.

$$GM: 2 < 5 \checkmark$$

$$GD: 1 < 2 \checkmark$$

$\therefore$ It is linked.



Ans

$P_1 = (0, 10)$	44	
$P_2 = (11, 11)$	46	Distance < 2
$P_3 = (12, 12)$	45	Direction $< 5^\circ$
$P_4 = (18, 18)$	92	TH
$P_5 = (19, 19)$	91	

e. Global edge linking:

$$P_{1,2} = 2, 3$$

$$P_{4,5} = 8, 9$$

$$\theta = 135^\circ$$

$$P_2 = 4, 5$$

$$P_5 = 5, 1$$

$$P_3 = 6, 7$$

$$\sigma = x \cos \theta + y \sin \theta$$

$$\begin{aligned} \text{i) } \sigma_1 &= 2 \cos 135^\circ + 3 \sin 135^\circ \\ &= 2 \sin 45^\circ + 3 \cos 45^\circ \\ &= \sqrt{2} + \frac{3}{\sqrt{2}} \\ &= 0.707 \end{aligned}$$

$$\begin{aligned} \text{ii) } \sigma_2 &= 4 \cos 135^\circ + 8 \sin 135^\circ \\ &= 0.707 \end{aligned}$$

$$\begin{aligned} \text{iii) } \sigma_3 &= 6 \cos 135^\circ + 7 \sin 135^\circ \\ &= 0.707 \end{aligned}$$

$$\begin{aligned} \text{iv) } \sigma_4 &= 8 \cos 135^\circ + 9 \sin 135^\circ \\ &= 0.707 \end{aligned}$$

$$\begin{aligned} \text{v) } \sigma_5 &= 2 \cos 135^\circ + 1 \sin 135^\circ \\ &= -2.828 \end{aligned}$$

Advantage;

- Detect long straight line accurately.
- suitable for land detection.

• can detect partially broken lines.

Understanding scenario (20%)	Identification of key issues (20%)	Application of concept (25%)	Decision making (20%)	clarity of response (15%)	Total (100%)
20	20	25	20	25	100